The Wayback Machine - https://web.archive.org/web/20220816103154/https://help.ftx.com/hc/en-us/articles/360027668712-Liquidations

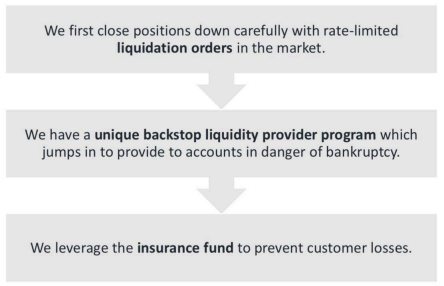
Liquidations

FTX Crypto Derivatives Exchange
Updated 12 hours ago

BROWSE

Notes:
1) While FTX aims to reduce the likelihood of clawbacks, they are still possible.
2) Estimated liquidation price is just an estimate. When an account actually gets liquidated will depend on multiple factors including the performance of all contracts your account holds a position in, the currency your collateral is in, and other factors.
3) FTX is not offered to US customers.
4) None of this is investment advice.

FTX significantly reduces the likelihood of clawbacks by using a three-tiered liquidation model:



Step 1:

An account begins to get liquidated if its margin fraction is less than its maintenance margin. So if its maintenance margin fraction is 3% then it would begin to get liquidated once it became 33x leveraged. During liquidation, users are unable to send orders on their account: the liquidation engine takes over.

The liquidation engine will then periodically send orders in the market to close down the account's position. The goal of the liquidation engine is to carefully close down positions in the market while minimizing impact, keeping markets orderly. The liquidation engine just sends standard limit orders on behalf of the account getting liquidated.

The speed of the liquidation will depend on the position size but for small positions it will aim to fully close down the position in about one minute. If partially liquidating the account causes its leverage to drop back below the threshold, the liquidation will end.

Step 2:

If account falls even closer to bankruptcy, the backstop liquidity provider system will kick in. This happens if the account's margin drops below the auto-close margin fraction. So if the auto-close margin fraction is 2%, then if the account becomes 50x leveraged, the account will begin to close down against the backstop liquidity providers.

When an account is getting auto-closed, it will have its position closed down at the bankruptcy price, and backstop liquidity providers will take over the position. A portion of the remaining collateral goes to the backstop liquidity fund.

Step 3:

If an account does go bankrupt, the backstop liquidity fund will pay out to bring the account's balance back to 0.

The more technical explanation:

Step 1:

If the account's margin fraction is less than maintenance margin but above auto-close margin fraction, then:

Approximately every 6 seconds, we send 10% of the position size as a order on the market, between 1 and 5 basis points through the book, with the constraint that the total sizes of liquidation orders is less than 0.0001 times the average daily volume of the underlying coin summed across all liquidating positions.

Specifically:

1. Every second, for each future, with probability ⅙:
    1. Set order size to 10% of position size
    2. Bound order notional from below by min($1000, position size)
    3. Bound order size from above by max liquidation size remaining
    4. Multiply order size by uniform(0.5, 1.5)
    5. Bound order size from above by position size
    6. Decrease max liquidation size remaining by order size
    7. Send order uniform(1bp, 5bp) through the book, expiring 1 second later
    1. Set max liquidation size remaining to 0.0001 times the underlying ADV (note: this is a global max shared between all accounts)
    2. For each account whose margin fraction is between maintenance margin and auto-close margin fraction, in random order:

Step 2:

If the account's margin fraction is less than auto-close margin fraction, then:

Every second, auto-close (1 - margin fraction / auto-close margin fraction) * position size, bounded below by min($1000, position size). If margin fraction < 0, auto-close the entire position.

Backstop Liquidity Providers (BLPs) have a max capacity per minute and per hour. Position is closed against BLPs in proportion to remaining capacity. If BLPs total capacity is insufficient, the remaining size is closed against users with large opposing positions (starting with the top 10 opposing positions, more if their total is insufficient), in proportion to their position sizes.

The positions on the liquidated account close at their PZP. The BLP takes over the account at ½ * PZP + ½ * MP, but not worse than MP plus MP * 10% * auto-close margin fraction. backstop liquidity fund covers the rest--i.e. it gets ½*abs(MP-PZP) if the account isn't yet bankrupt, and pays abs(MP-PZP) + 0.1 * MP * ACMF if it is. If account is bankrupt and backstop liquidity fund is empty, the remaining is taken from positions with positive unrealized pnl (proportionally to pnl).

BLP auto-closing orders do not print in the trade history.

If a contract hits a circuit breaker, MP is the premium as of when the circuit breaker was enacted plus current index price.

Step 3:

Whenever an account hits the auto-close margin fraction, the backstop liquidity fund steps in. If the account isn't yet bankrupt, then the backstop liquidity fund **gains** 1/3 of the remaining value of the account. If the account *is* already bankrupt, then the backstop liquidity fund **pays** the negative value of the account (plus 10% of the auto-close margin fraction) to prevent clawbacks.

Customers will only have their positions auto-closed if an account hits auto-close margin fraction *and* the backstop liquidity providers are out of capacity. Customers only face clawbacks if an account goes bankrupt *and* the backstop liquidity fund is empty.

How can I determine my liquidation risk?

There are two ways to determine when your account would get liquidated.

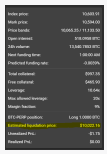
#1: Estimated Liquidation Price

If you go to any market page there will be an informational box on the right hand side. You can find an *Estimated liquidation price* there; if you don't have any other positions on then your position will start to get liquidated if the future's mark price gets there.

ARTICLES IN THIS SECTION
What Are Futures?
Account Margin Management
Collateral Management
Complete Futures Specs
Settlement & Delivery
Funding
Contracts Specs
Order Limits and Price Bands
Liquidations
Backstop Liquidity Provider Program

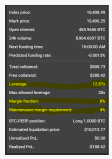
RELATED ARTICLES
Account Margin Management
Complete Futures Specs
Fees
Spot Margin Trading Explainer
Order Limits and Price Bands

PROMOTED ARTICLES
FTX Features Overview
Introducing New Fee and Ratelimit Tiers to the FTX VIP & Market Maker Program
Deposit to your FTX account to earn VIP status
VIP Program and Market Maker Policy
FTX partners with Paradigm for one-click futures spreads trading
Ratelimits on FTX



#2: Maintenance Margin Requirement

Once again look at the informational box on the right hand side. You can also find your current amount of leverage used there, and your current m*argin fraction*, which is just 1/leverage. Your account will begin to get liquidated if your margin fraction drops below the *maintenance margin requirement*, also displayed in the box. Maintenance margin fraction starts at 3% and increases with position size, so you will begin to get liquidated if your account gets to ~33x leverage (or less depending on position size).



In the example above the user has a margin fraction of 8%:

Position size = 1 BTC * 10,406.25 $/BTC = $10,406

Total collateral = $808.73

Margin fraction = 808.73/10406.25 = 8%.

They will get liquidated if their margin drops down to the maintenance margin requirement of 4%. That means they'll get liquidated if markets move 8% - 4% = 4% down.

In general, the formula for estimated liquidation price should be:

Estimated Liquidation Price = Mark Price * (1 + Maintenance margin requirement - Total Collateral / Total Notional Size) if you are long

Estimated Liquidation Price = Mark Price * (1 - Maintenance margin requirement + Total Collateral / Total Notional Size) if you are short

Note that Total Notional Size here refers to the sum of your open position size, measured in USD, across all positions in that subaccount. Also note that estimated liquidation prices can be unintuitive, especially if you have positions in multiple contracts. **Depend on the relative performance of different positions you hold, the actual price your account would get liquidated at might be different from your estimated liquidation price**: it's impossible to show a perfect estimated liquidation price for BTC-PERP if you also have an XRP-PERP position on, because your account's margin will also depend on the performance of XRP-PERP.

Note that we try to liquidate accounts slowly, and we will stop if your account's margin fraction gets above maintenance, so we might only have to liquidate part of your position. For more information see here.

If you are worried about liquidations and don't want to actively manage your margin, consider trading leveraged tokens; they allow 3x leverage and automatically rebalance to avoid liquidations. For more information see here.

Collateral

Collateral for the futures is in stablecoins. The current set of accepted stablecoins is USDC, TUSD, USDP, BUSD.

To deposit or withdraw collateral, go to your wallet page and deposit either USDC, TUSD, USDP. Depositing either will credit your account with 'USD', which is automatically used as collateral for all of your futures trades.

By default all margin is posted in 'USD' in your wallet. USD can be funded by depositing USDC, TUSD, USDP, BUSD.

See Non-USD Collateral for more details.

Any positive PnL will be paid out in USD regardless of which type of collateral you use.See here for more details.

By default all positions use the same collateral pool, and all USD, non-USD fiat, and above cryptocurrencies in your wallet count as collateral. Each subaccount has one central collateral wallet and uses cross margining for the account. Each subaccount has separate margin and collateral from other subaccounts.

If you want to use isolated margin create a subaccount for that position and move in collateral.

← Previous
Order Limits and Price Bands

Next →
Backstop Liquidity Provider Program

Didn't find what you were looking for?
Create a support ticket

Community

Was this article helpful?
Yes No
1 out of 1 found this helpful

FTX
FTX Services and FTX Token (FTT) are not available in the United States or other prohibited jurisdictions
English (US)